

Ex. No.: 5

Date:

System Calls Programming

Aim: To experiment system calls using fork(), execlp() and pid() functions.

Algorithm:

1. **Start**
 - Include the required header files (stdio.h and stdlib.h).
2. **Variable Declaration**
 - Declare an integer variable pid to hold the process ID.
3. **Create a Process**
 - Call the fork() function to create a new process. Store the return value in the pid variable:
 - If fork() returns:
 - -1: Forking failed (child process not created).
 - 0: Process is the child process.
 - Positive integer: Process is the parent process.
4. **Print Statement Executed Twice**
 - Print the statement:

scss

Copy code

THIS LINE EXECUTED TWICE

(This line is executed by both parent and child processes after fork()).

```

1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <unistd.h>
4 int main(){
5     int pid;
6     pid = fork();
7     printf("This line executed twice\n");
8     if(pid == -1){
9         printf("Child process not created\n");
10        exit(0);
11    }
12    else if(pid == 0){
13        printf("\n--- Child Process ---\n");
14        printf("Child process Id (PID): %d\n", getpid());
15        printf("Parent process ID (PPID): %d\n",getppid());
16        printf("Executing 'ls' command using execlp()\n");
17        execlp("ls", "ls", NULL);
18    }
19    else{
20        printf("\n--- Parent Process ---\n");
21        printf("Parent Process Id (PID): %d\n", getpid());
22        printf("Parent's parent process Id (PPID): %d\n", getppid());
23    }
24    printf("It can be executed twice\n");
25    return 0;
26 }

```

[elamathi_066@localhost ~]\$ gcc format_program.c -o format_program

[elamathi_066@localhost ~]\$./format_program.c

bash: ./format_program.c: Permission denied

[elamathi_066@localhost ~]\$./format_program

This line executed twice

--- Parent Process ---

Parent Process Id (PID): 3408

Parent's Parent Process Id (PPID): 2508

It can be executed twice

This line executed twice

--- child Process ---

Child Process Id (PID): 3409

Parent Process ID (PPID): 1887

Executing 'ls' command using execlp()

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[elamathi_066@localhost ~]\$